

Element H

STEM Principles

*Note: scoring rubric for these tests is found in Element C

Design Requirements (In priority order (high-low))	Plan of Testing
Produce Sufficient Water Flow	To accurately measure water flow, we would take a marked container (a beaker, or something we know the volume of) and time how long it takes to fill up said container. Divide 60 by the number of seconds it took to fill the container. This gives you the flow rate in gallons per minute. This is really the only way to scientifically test our flow rate, however, we do also plan to test how different flow rates would affect the students in the classroom. For the activities being executed by these students, who may not know how to use a sink, a lower flow rate may be more effective. Additionally, we plan on receiving feedback of the overall design via email from the teachers in Whitemarsh Elementary's Life Skills class, which could also give us more information on how effective this flow rate is for the students. (Repeat 10 times to mitigate error)
Portability	To accurately measure the ability for a person to move the sink, we would measure the sink in its entirety in pounds on a scale. This would allow us to gauge the portability based off of the United States Department of Labor, in order to be successful we would need to be 51 lbs or below. Another way we plan to test the portability is by having multiple tests on the carrying ability of the sink from its handles with different groups of people to accomplish this we will make sure to sample from people of different size and motor skill abilities. We will test our design by weighing it and utilizing the feedback from the people who test the handles. With this our goal is create a sink that has balance and is usable by the teachers at Whitemarsh Elementary without strain or difficulty.
Student Functionality	In order to create a sink that is usable by the life skills class at Whitemarsh, it is fundamental to our

	design that we are knowledgeable about the needs of the students. If the students cannot operate the sink, then it has failed to do its job properly. In order to test the student functionality we will conduct an experiment consisting of different people trying to use the sink with minimal background knowledge given to them. In order to best emulate the situation the sink would be in, we would utilize people with similar motor skills to best test the ease of use for students in the life skills class at Whitemarsh Elementary. To be successful with our sink we would like at least 80% of students who attempt to use the sink to be able to use it without much struggle. The main goal of this test is to make sure that our sink can accommodate those with lesser developed motor skills. (All students should be accounted for)
Standard Sink Appearance	Through our research, we learned the importance realism plays in the classroom, especially for special needs students. That is why this sink was made with realism in mind; so that these students will someday be able to go out into the real world and use a sink properly. We did our best with what we had to create a sink that resembled that of a real world sink. We included a mirror on one of our structures to give it that realistic feel. In addition, there are large handles with an “H” and a “C” to resemble how a real sink has hot and cold water, and a large sink bowl. All of these aspects were carefully chosen to give the children an understanding of what any given sink could look like. Ultimately, realisticness can only be measured by how you look at something. We can measure the appearance of the sink by having random people look at the design and try to figure out what it is. Also, when we receive our stakeholder feedback, we can ask for an assessment on how realistic our sink looks.
Safety	Safety is important when it comes to a life skills class, as with students who are not as developed some things people think of as common knowledge does not come as easily to them. To make sure our design fits our safety requirement we will have to fully check in and around the sink to make sure there are no sharp edges or anything dangerous hanging out. We all would need to make sure all of our parts are put together in a way in which it is difficult for them to come apart

	easily to ensure no kids can easily knock it down or break it apart exposing sharp edges. Also in the testing of safety we would incorporate the weight of the sink, as it has to be light enough for the teacher to pick up without injury and also light enough for the students to not get seriously injured if the sink were to hit a students hands or feet. Our goal with safety is to create a sink that will not cause physical issues with the movement or use of the sink resulting in an injury.
Cleanliness	Another prominent issue in any classroom is the cleanliness of it, especially in an elementary school. In the Life Skills classroom, the students learn differently, so things need to be very clean and sterile at all times to allow for the student's learning needs. That is why cleanliness was an extremely important part of this design process. We ensured that the entire design was sturdy enough to withstand any outstanding pressure, but could also be easily taken apart for cleaning purposes. The tubes that connect to our makeshift faucet can be disconnected and cleaned very easily, the two gallon jugs can be removed easily as well, and the entire top of the design can be taken off to allow a deep clean of the bottom storage container which will hold all of the wastewater. We can measure cleanliness by how easy it is to clean this design and how long it can go without being cleaned. This too will be a part of our feedback via email from the Life Skills classroom. (Repeat for each week for a one month period)
Storage	Storage is a big issue for this classroom, as they are very tight for space as it is. We wanted to create something that could be easily stored for the teachers to grab and go, and also so the students are not always distracted by a big sink in the middle of the room. We can measure storage through the overall size of the sink, how much physical space does it take up(volume cubic ft)? Along with everything else, the feedback we get from the teachers in the Life Skills class is the main way we will be able to know the overall effectiveness of our design in this particular area.
Customizability	Customizability refers to the addition and accessibility that the teacher has access to on the sink other than the main functionality of flowing

	water. In order to test this we will count the amount of additional features added with 3 or more being excellent. We will also compare the features added to how they naturally work when not put onto a portable sink. To help measure that we will collect data from different users about how the customized tools affect their use of the sink and if they are helpful to the learning of someone using how to use a sink. Our main objective with this testing is to ensure that the space we are taking up on the sink is worth it and uses the space efficiently while producing results similar to a regular sink.
--	--

Procedure Feedback from James Muscarella and Drexel Professors Feedback (via zoom):

Red: Feedback (Both experts condensed due to similar feedback)

Green: Clarification

Produce Sufficient Water Flow: None'

Portability: This accounts for weight, but portability also involves shape.- An item can be cumbersome without necessarily being too heavy.

To clarify this issue, we acknowledge that our testing of portability was solely based on weight. However, we understand that it is difficult to mathematically measure this requirement. Further, to add depth to this test, we have discussed the possibility of sampling a random sample of students within our highschool, and have them carry the sink from one room to another, and then provide a rating from 1-10 to rate the difficulty of the action. This would provide greater depth to the test. Due to lack of time, this procedure has been halted and will be proceeded with if time permits.

Student Functionality: The idea of this test is very general. What specifically can be done to test a range of students?

To further this thought, to test a range of students we would measure the amount of students who were able to successfully turn on or turn off the sink under a 5 minute time. While we understand all students within these classrooms have varying capabilities, we understand that these times may vary due to those factors. However, our goal is to make this sink be used by all students within the classroom with ease, so

if it takes a student longer than a 5-minute span to use the sink properly, we would deem that student's performance a failure, lowering the % of students in the classroom who can successfully use the sink.

Standard Sink Appearance: None

Safety: **What safety parameters will be tested that relate to the students?**

Safety parameters that will be tested that relate to the students safety is the number of reports the teachers make involving scrapes, cuts, etc that were resulted from sink use. We will gather these results after one week of use. A report of 2 injuries or more would be deemed unsafe, and would further force the group to recollect the prototype. While this is not on our scoring rubric in terms of the number of reported injuries, the results of this report would further lead the group to have either recollect the prototype, or remain confident that this was not a necessary occurrence to have to be concerned about for future use.

Cleanliness: **How will you know something is 'clean'? Appearance? Swab for bacteria?**

While we understand that appearance does not always determine cleanliness, for this project, we are strictly looking for any visible patches of dirt, residues, stains, etc. We understand the bacterial possibility, but believe that with the proper instruction to clean the sink with clorox wipes every 2 weeks, bacterial presence should remain low and this should not be a major concern needing to be measured.

Storage: **Is there a procedure to 'pack up' or condense the sink to its smallest configuration?**

Yes! We have made a video for the stakeholders as well as our experts to explain to them how to use the sink, as well as how to properly pack up the sink in a configuration that maximizes the sink's ability to fit away in a confined space.